

LCM and HCF

Factors and Multiples: If a divides b exactly, we say that a is a factor of b and also we say that b is multiple of a .

i.e. 7 is a factor of 14, 8 is factor of 24 e.t.c. or 14 is multiple of 7, 24 is multiple of 8.

HCF/G.C.D/G.C.M: The HCF of two or more than two numbers is the greatest number that divides each of them exactly.

Method of finding HCF

- (i) **Factorization method:** Express each one of the given numbers as the product of prime factors. The product of common prime factors with least power gives HCF.

Example: Find the HCF of 42, 63 and 140, $42 = 7 \times 2 \times 3$, $63 = 7 \times 3 \times 3$, $140 = 7 \times 5 \times 2 \times 2$, So HCF = 7

- (ii) **Division Method:** Suppose we have find the HCF of two given numbers. Divide the larger number by the smaller one. Now divide the divisor by the remainder . Repeat this process till remainder is zero. The last divisor is required HCF.

Example: Find the HCF of 148 and 185

$$\begin{array}{r} 148 \overline{)185} (1 \\ \underline{148} \\ 37 \overline{)148} (4 \\ \underline{148} \\ \hline \end{array}$$

i.e. HCF = 37.

LCM : The least number which is exactly divisible by each one of the given number is called their LCM.

Methods

- (1) **Factorization method:** Resolve each of the given number into a product of prime factors. LCM is the product of terms of highest power of all factors.

Product of two numbers = their LCM $\times$ their HCF

Co-prime: Two numbers are said to be co-prime if their HCF is 1.

i.e. 4 and 3 are co-prime numbers.

HCF and LCM of fractions

$$(i) \text{ HCF} = \frac{\text{HCF of numerators}}{\text{LCM of denominators}}$$

$$(ii) \text{ LCM} = \frac{\text{LCM of numerators}}{\text{HCF of denominators}}$$

Important Results

- (i) Product of two numbers = HCF of the numbers $\times$ LCM of the numbers
- (ii) The greatest number which divides the number x , y and z leaving remainders a , b and c respectively.
= HCF of $(x - a)$ $(y - b)$ $(z - c)$
- (iii) The least number which when divided by x , y and z leaves the remainder a , b and c respectively, is given by
[LCM of $(x, y, z) + K$], where $K = (x - a) = (y - b) = (z - c)$
- (iv) The least number which divided by x , y and z leaves the same remainder k in each case, is given by
[LCM of $(x, y, z) + K]$

- (v) The greatest number that will divide x , y and z leaving the same remainder in each case, is given by
 $[\text{HCF of } (x - y), (y - z), (z - x)]$
- (vi) When the HCF of each pair of n given numbers is a and their LCM is b , then product of these numbers is given by $(a)^{n-1} \times b$ or $(\text{HCF})^{n-1} \times \text{LCM}$

Types of Questions

1. HCF of $\frac{2}{3}$, $\frac{4}{5}$ and $\frac{6}{7}$

Sol. HCF of fractions = $\frac{\text{HCF of numerators}}{\text{LCM of denominators}}$

$$= \frac{\text{HCF}(2, 4, 6)}{\text{LCM}(3, 5, 7)} = \frac{2}{105}$$

2. The HCF (GCD) of a , b is 12 and a and b are positive integers and $a > b > 12$. The smallest value of (a, b) are respectively

Sol. Given HCF of $a, b = 12$

Let the numbers be $12x$ and $12y$, where x and y are co-prime.

But given $a > b > 12$

i.e. $a = 36$

$b = 24$

3. The LCM of three different numbers is 120. Which of the following cannot be HCF

- (i) 4 (ii) 12 (iii) 35 (iv) 8

Sol. LCM of three no is = 120

Now, factors of $120 = 2 \times 2 \times 2 \times 3 \times 5$,

Hence HCF can be 4, 8, 12

But 35 can't be HCF.

4. Two numbers are in the ratio 3 : 4. If their LCM is 84, then the greater number is

Sol. Let the number be $3x$ and $4x$

$\text{LCM of } 3x \text{ and } 4x = 12x$

$\text{LCM} = 84$

$12x = 84$

$x = 7$

Greatest Number = $4 \times 7 = 28$

5. A rectangular piece of cloth has dimensions 16 m and 12m. What is the least number of equal square that can be cut out of this cloth?

Sol. HCF of 16 and 12 = 4

No of pieces = $\frac{16 \times 12}{4 \times 4} = 3 \times 4 = 12$

6. Find the largest number, which divides 34, 90, 104, leaving the same remainder in each case.

Sol. Difference between numbers = $90 - 34$, $104 - 90$

and $104 - 134$

= 56, 14 and 70.

HCF of 56, 14 and 70 = 14

i.e. 14 is largest number.

7. Which is the smallest multiple of 7, which when divided by 6, 9, 15 and 18 respectively leaves 4 as remainder in each case.

Sol. LCM of 6, 9, 15 and 18 = 90

Remainder = 4, but number is also divisible by 7 so required number

= $90k + 4$... (i)

By putting

$k = 1, 2, 3, 4, \dots$ and checking if it is divisible by 7

Required number = $90k + 4 = 90 \times 4 + 4 = 364$

Foundation

Questions

- LCM of $\frac{2}{3}, \frac{4}{9}, \frac{5}{6}$ is
 - $\frac{20}{3}$
 - $\frac{10}{3}$
 - $\frac{20}{27}$
 - $\frac{8}{27}$
- The HCF of two numbers is 8. Which one of the following can never be their LCM?
 - 24
 - 48
 - 56
 - 60
- The LCM of two numbers is 520 and their HCF is 4. If one of the numbers is 52, then the other number is
 - 40
 - 42
 - 50
 - 52
- The HCF of two numbers is 96 and their LCM is 1296. If one of the numbers is 864, the other is
 - 132
 - 135
 - 140
 - 144
- The product of two numbers is 216. If the HCF is 6, then their LCM is
 - 72
 - 60
 - 48
 - 36
- Two numbers are in the ratio 3 : 4. If their HCF is 4, then their LCM is
 - 48
 - 42
 - 36
 - 24
- The LCM of two numbers, which are multiples of 12 is 1056. If one of the numbers is 132, the other number is
 - 12
 - 72
 - 96
 - 132
- Two numbers are in the ratio 3 : 4. The product of their HCF and LCM is 2028. The sum of the numbers is
 - 68
 - 72
 - 86
 - 91
- Two numbers are in the ratio 3 : 4. If their LCM is 240, the smaller of the two number is
 - 100
 - 80
 - 60
 - 50
- The HCF of two numbers is 16 and their LCM is 160. If one of the number is 32, then the other number is
 - 48
 - 80
 - 96
 - 112
- The product of two numbers is 4107. If the HCF of the numbers is 37, the greater number is
 - 185
 - 111
 - 107
 - 101
- The ratio of two numbers is 3 : 4 and their HCF is 5. Their LCM is
 - 80
 - 48
 - 120
 - 60
- The LCM of two numbers is 1820 and their HCF is 26. If one number is 130. Then, the other number is
 - 70
 - 1690
 - 364
 - 1264
- The HCF of two numbers 12906 and 14818 is 478. Their LCM is
 - 400086
 - 200043
 - 60012
 - 800172
- What is the greatest number which will divide 110 and 128 leaving a remainder 2 in each case?
 - 8
 - 18
 - 28
 - 38
- The smallest perfect square divisible by each of 6, 12 and 18 is
 - 196
 - 144
 - 108
 - 36
- When a number is divided by 15, 20 and 35, each time the remainder is 8. Then, that smallest number is
 - 428
 - 427
 - 328
 - 388
- Find the HCF of 35 and 30.
 - 5
 - 6
 - 7
 - 8
- The HCF and the product of two numbers are 15 and 6300 respectively. The number of possible pairs of the numbers are.
 - 4
 - 3
 - 2
 - 1
- The LCM of two numbers is 12 times their HCF. The sum of the HCF and the LCM is 403. If one of the numbers is 93, then other number is
 - 124
 - 128
 - 134
 - 138
- Find the least number exactly divisible by 12, 15, 20, 27.
 - 450
 - 540
 - 230
 - 640

22. The largest number which divides 25, 73 and 97 to leave the same remainder in each case, is:
 (a) 24 (b) 23
 (c) 21 (d) 6
23. Find the largest 5 digits number which is exactly divisible by 12, 15, 18, 27.
 (a) 90000 (b) 99999
 (c) 99010 (d) 99900
24. Find the least number which when divided by 20, 25, 35, 40 leaves remainders 14, 19, 29 and 34 respectively.
 (a) 1220 (b) 1394
 (c) 1365 (d) 1470
25. 3 different containers contain 496l, 403l and 713l mixture of milk and water. What biggest measure of a container can measure all the 3 quantities exactly?.
 (a) 31 l (b) 41 l
 (c) 51 l (d) 52 l
26. Three bells ring simultaneously at 11am. They ring at regular intervals of 20 min, 30 min and 40 min, respectively. The time when all the three ring together next is:
 (a) 2 pm (b) 1 pm
 (c) 1:15 pm (d) 1:30 pm
27. Four runners started running simultaneously from a point on a circular track. They took 200 s, 300 s, 360 s and 450 s to complete one round. After how much time do they meet at the starting point for the first time?
 (a) 1800 s (b) 3600 s
 (c) 2400 s (d) 4800 s
28. The LCM of two positive integers is twice the larger number. The difference of the smaller number and the HCF of the two numbers is 4. The smaller number is:
 (a) 12 (b) 6
 (c) 8 (d) 10
29. Six bells commence tolling together at intervals of 2, 4, 6, 8, 10 and 12 s, respectively. In 30 min., how many times do they toll together?
 (a) 16 (b) 15
 (c) 10 (d) 4
30. Rajesh is incharge of buying bread rolls and buns for a party. There are 10 buns in each box of buns and 8 bread rolls in each box of bread rolls. Rajesh wants to buy exactly the same number of buns and bread rolls. What is the smallest number of boxes he should buy for buns alone?
 (a) 10 (b) 8
 (c) 4 (d) 5

Moderate

1. Three tankers contain 403 l, 434 l, 465 l of diesel, respectively. Then, the maximum capacity of container that can measure the diesel of the three containers in exact number of times is
 (a) 31 l (b) 62 l
 (c) 41 l (d) 84 l
2. A, B and C start together from the same point to travel around a circular path of 30 km in circumference. A and B are travelling in the same direction and C in the opposite direction. If A travels 5 km, B travels 7 km and C travels 8 km in an hour, then they all come together again after:
 (a) 25 h (b) 30 h
 (c) 15 h (d) 20 h
3. The traffic lights at three different road crossings change after 24 s, 36 s and 54 s, respectively. If they all change simultaneously at 10 : 15 : 00 am, then at what time will they again change simultaneously?
 (a) 10 : 16 : 54 am (b) 10 : 18 : 36 am
 (c) 10 : 17 : 02 am (d) 10 : 22 : 12 am
4. Sum of two numbers is 384, HCF of the numbers is 48. The difference of the number is
 (a) 100 (b) 192
 (c) 288 (d) 336
5. The sum of two numbers is 45. Their difference is $\frac{1}{9}$ of their sum. Their LCM is
 (a) 200 (b) 250
 (c) 100 (d) 150
6. The ratio of the sum to the LCM of two natural numbers is 7 : 12. If their HCF is 4, then the smaller number is
 (a) 20 (b) 16
 (c) 12 (d) 8
7. If the students of a class can be grouped exactly into 6 or 8 or 10, then the minimum number of students in the class must be
 (a) 60 (b) 120
 (c) 180 (d) 240
8. Three numbers are in the ratio 2 : 3 : 4 and their HCF is 12. The LCM of the numbers is
 (a) 144 (b) 192
 (c) 96 (d) 72
9. The greatest common divisor of $3^{333} + 1$ and $3^{334} + 1$ is
 (a) $3^{333} + 1$ (b) 20
 (c) 2 (d) 1

10. A milk vendor has 21 l of cow milk, 42 l of toned milk and 63 l of double toned milk. If he wants to pack them in cans, so that each can contains same number of litres of milk and does not want to mix any two kinds of milk in a can, then the least number of cans required as
 (a) 3 (b) 6
 (c) 9 (d) 12
11. There are 24 peaches, 36 apricots and 60 bananas and they have to be arranged in several rows in such a way that every row contains the same number of fruits of only one type. What is the minimum number of rows required for this to happen?
 (a) 12 (b) 9
 (c) 10 (d) 6
12. Three numbers which are coprimes to one another are such that the product of the first two is 551 and that of the last two is 1073. The sum of the three numbers is
 (a) 75 (b) 81
 (c) 85 (d) 89
13. Three sets of English, Mathematics and Science books containing 336, 240 and 96 books, respectively have to be stacked in such a way that all the books are stored subject-wise and the height of each stack is the same. Total number of stacks will be
 (a) 14 (b) 21
 (c) 22 (d) 48
14. The sum of two numbers is 36 and their HCF and LCM are 3 and 105, respectively. The sum of the reciprocals of two numbers is
 (a) $\frac{2}{35}$ (b) $\frac{3}{25}$
 (c) $\frac{12}{35}$ (d) $\frac{2}{25}$
15. What is the least number of square tiles required to pave the floor of a room 15 m 17 cm long and 9 m 2 cm broad?
 (a) 840 (b) 841
 (c) 820 (d) 814
16. The HCF (GCD) of a, b is 12. a, b are positive integers and $a > b > 12$. The smallest values of (a, b) are respectively.
 (a) 12, 24 (b) 24, 12
 (c) 24, 36 (d) 36, 24
17. If $P = 2^3 \times 3^{10} \times 5$ and $Q = 2^5 \times 3 \times 7$, then HCF of P and Q is
 (a) $2 \cdot 3 \cdot 5 \cdot 7$ (b) $3 \cdot 2^3$
 (c) $2^2 \cdot 3^7$ (d) $2^5 \cdot 3^{10} \cdot 5 \cdot 7$
18. The sum of two numbers is 84 and their HCF is 12. Total number of such pairs of numbers is
 (a) 2 (b) 3
 (c) 4 (d) 5
19. The sum of two numbers is 36 and their HCF is 4. How many pairs of such numbers are possible?
 (a) 1 (b) 2
 (c) 3 (d) 4
20. The HCF of two numbers 12908 and 14808 is 672. Their LCM is
 (a) 284437 (b) 200043
 (c) 60012 (d) 800172
21. Two jars of capacity 50 l and 80 l are filled with oil. What must be the capacity of a mug that can completely measure the oil of the two jars?
 (a) 5 l (b) 15 l
 (c) 10 l (d) 20 l
22. The traffic lights at three different road crossings change after every 48 sec, 72 sec, and 108 sec respectively. If they all change simultaneously at 8 : 20 hours, then at what time will they again change simultaneously?
 (a) 8 : 27 : 12 hours (b) 8 : 25 : 14 hours
 (c) 8 : 24 : 12 hours (d) 8 : 29 : 12 hours
23. The LCM of two numbers is 2310 and their HCF is 30. If one of the number is 7×30 , Find the other number.
 (a) 320 (b) 330
 (c) 340 (d) 350
24. What is the least multiple of 7, which when divided by 2, 3, 4, 5 and 6 leaves the remainders 1, 2, 3, 4 and 5 respectively?
 (a) 117 (b) 119
 (c) 121 (d) 123
25. The least number, which when divided by 12, 15, 20 or 54 leaves a remainder 4 in each case is:
 (a) 450 (b) 454
 (c) 540 (d) 544
26. The maximum number of students among whom 1001 pens and 910 pencils can be distributed in such a way that each student gets same number of pens and same number of pencils, is
 (a) 91 (b) 910
 (c) 1001 (d) 1911
27. A, B and C start running at the same time and from the same point in the same direction in a circular stadium. A completes a round in 252 s, B in 308 s and C in 198s. After what time will they meet again at the starting point?
 (a) 26 min 18 s (b) 42 min 36 s
 (c) 45 min (d) 46 min 12 s
28. Three men step-off together from the same spot. Their steps measures 63 cm, 70 cm and 77 cm, respectively. The minimum distance each should cover, so that all can cover the distance in complete steps, is
 (a) 9630 cm (b) 9360 cm
 (c) 6930 cm (d) 6950 cm

29. Find the largest number of four digits such that on dividing by 15, 18, 21 and 24 the remainders are 11, 14, 17 and 20, respectively.
 (a) 6557 (b) 7556
 (c) 5675 (d) 7664
30. The total number of integers between 100 and 200, which are divisible by both 9 and 6 is
 (a) 5 (b) 6
 (c) 7 (d) 8

Difficult

1. The largest possible length of a tape which can measure 525 cm, 1050 cm and 1155 cm length of cloths in a minimum number of attempts without measuring the length of a fraction of the tape's length is
 (a) 25 (b) 105
 (c) 75 (d) None of these
2. There are three drums with 1653 litre, 2261 litre and 2527 litre of petrol. The greatest possible size of the measuring vessel with which we can measure up the petrol of any drum while every time the vessel must be completely filled is:
 (a) 31 (b) 27
 (c) 19 (d) 41
3. Mr. Baghwan wants to plant 36 mango trees, 144 orange trees and 234 apple trees in his garden. If he wants to plant the equal no. of trees in every row, but the rows of mango, orange and apple trees will be separate, then the minimum number of rows in his garden is :
 (a) 18 (b) 23
 (c) 36 (d) Can't be determined
4. Find the least possible perfect square number which is exactly divisible by 6, 40, 49 and 75
 (a) 176400 (b) 15000
 (c) 175600 (d) 16500
5. Three bells in the Bhootnath temple toll at the interval of 48, 72 and 108 seconds individually. If they have tolled all together at 6 : 00 AM then at what time will they toll together after 6 : 00 AM?
 (a) 6 : 07 : 15 AM (b) 6 : 07 : 12 AM
 (c) 4 : 04 : 12 AM (d) 6 : 06 : 12 AM
6. What is the least possible number which when divided by 18, 35 and 42 leaves 2, 19 and 26 as the remainders respectively?
 (a) 400 (b) 740
 (c) 614 (d) 621
7. Find the HCF of 0.0005, 0.005, 0.15, 0.175, 0.5 and 3.5.
 (a) .0005 (b) .005
 (c) .05 (d) .5
8. The least number which when divided by 2, 3, 4, 5 and 6 leaves the remainder 1 in each case. If the same number is divided by 7 it leaves no remainder. The number is:
9. Three bells, toll at interval of 36 sec, 40 sec and 48 sec respectively. They start ringing together at particular time. They next toll together after :
 (a) 6 minutes (b) 12 minutes
 (c) 18 minutes (d) 24 minutes
10. Mr. Black has three kinds of wine, of the first kind 403 litres, of the second 434 litres and of the third 465 litres. What is the least number of full corks of equal size in which these can be stored without mixing?
 (a) 31 (b) 39
 (c) 42 (d) 51
11. Abhishek, Bobby and Charlie start from the same point and travel in the same direction round an Island 6 km in circumference. Abhishek travels at the rate of 3km/hr, Bobby at the rate of $2\frac{1}{2}$ km/hr and Charlie at the rate of $1\frac{1}{4}$ km/hour. In how many hours will they come together again?
 (a) 6 hrs (b) 12 hrs
 (c) 24 hrs (d) 15 hrs
12. The LCM of two numbers is 4 times their HCF. The sum of LCM and HCF is 125. If one of the numbers is 100, then the other number is
 (a) 5 (b) 25
 (c) 100 (d) 125
13. LCM of two numbers is 120 and their HCF is 10. Which of the following can be the sum of those two numbers?
 (a) 140 (b) 80
 (c) 60 (d) 70
14. A fraction becomes $\frac{1}{6}$ when 4 is subtracted from its numerator and 1 is added to its denominator. If 2 and 1 are respectively added to its numerator and the denominator, it becomes $\frac{1}{3}$. Then, the LCM of the numerator and denominator of the said fraction, must be

- (a) 14 (b) 350 (a) 70 (b) 77
(c) 5 (d) 70 (c) 63 (d) 56
15. From a point on a circular track 5 km long, A, B and C started running in the same direction at the same time with speeds of $2\frac{1}{2}$ km/h, 3 km/h and 2 km/h, respectively. Then, on the starting point all three will meet again after:
(a) 30 h (b) 6 h
(c) 10 h (d) 15 h
16. The HCF of two numbers is 15 and their LCM is 300. If one of the numbers is 60, the other is
(a) 50 (b) 75
(c) 65 (d) 100
17. HCF and LCM of two numbers are 7 and 140, respectively. If the numbers are between 20 and 45, the sum of the numbers is
(a) 99279 (b) 99370
(c) 99269 (d) 99530
18. The number of integers in between 100 and 600, which are divisible by 4 and 6 both is
(a) 40 (b) 42
(c) 41 (d) 50
19. The least number, which when divided by 18, 27 and 36 separately leaves remainders 5, 14 and 23 respectively, is
(a) 95 (b) 113
(c) 149 (d) 77
20. The largest number of five digits which, when divided by 16, 24, 30, and 36 leaves the same remainder 10 in each case, is
(a) 99279 (b) 99370
(c) 99269 (d) 99530

Previous Year (Memory Based)

1. The HCF and LCM of two numbers are 44 and 264 respectively. If the first number is divided by 2, then quotient is 44. The other number is
(a) 147 (b) 528
(c) 132 (d) 264
2. The ratio of two numbers is 5 : 6 and their LCM is 480, then their HCF is
(a) 20 (b) 16
(c) 6 (d) 5
3. Product of two coprime numbers is 117, then their LCM is
(a) 9 (b) 13
(c) 39 (d) 117
4. The LCM of two numbers is 48. The numbers are in the ratio 2 : 3. The sum of the numbers is
(a) 28 (b) 32
(c) 40 (d) 64
5. The ratio of two numbers is 4 : 5 and their HCF is 8. Then, their LCM is
(a) 130 (b) 140
(c) 150 (d) 160
6. If the HCF and LCM of two consecutive (positive) even numbers is 2 and 84 respectively, then the sum of the numbers is
(a) 30 (b) 26
(c) 14 (d) 34
7. The HCF and LCM of two numbers are 8 and 48 respectively. If one of the numbers is 24, then the other number is
(a) 48 (b) 36
(c) 24 (d) 16
8. The product of the LCM and the HCF of two numbers is 24. If the difference of the numbers is 2, then the greater of the number is
(a) 3 (b) 4
(c) 6 (d) 8
9. The HCF and product of two numbers are 15 and 6300, respectively. The number of possible pairs of the numbers is
(a) 4 (b) 3
(c) 2 (d) 1
10. The LCM of two numbers is 20 times their HCF. The sum of HCF and LCM is 2520. If one of the numbers is 480, the other number is
(a) 400 (b) 480
(c) 520 (d) 600
11. HCF of $\frac{2}{3}$, $\frac{4}{5}$ and $\frac{6}{7}$ is
(a) $\frac{48}{105}$ (b) $\frac{2}{105}$
(c) $\frac{1}{105}$ (d) $\frac{24}{105}$
12. The LCM and the HCF of the numbers 28 and 42 are in the ratio
(a) 6 : 1 (b) 2 : 3
(c) 3 : 2 (d) 7 : 2

13. If the LCM and HCF of two expressions are $(x^2 + 6x + 8)$ $(x + 1)$ and $(x + 1)$ respectively and one of the expression is $x^2 + 3x + 2$, then find the other.
 (a) $x^2 + 5x + 4$ (b) $x^2 - 5x + 4$
 (c) $x^2 + 4x + 5$ (d) $x^2 - 4x + 5$
14. Two numbers are in the ratio 3 : 4. Their LCM is 84. Then, the greater number is
 (a) 21 (b) 24
 (c) 28 (d) 84
15. Two numbers, both greater than 29, have HCF 29 and LCM 4147. The sum of the numbers is
 (a) 966 (b) 696
 (c) 669 (d) 666
16. The HCF and LCM of two numbers are 11 and 385, respectively. The numbers are
 (a) 55 and 57 (b) 55 and 77
 (c) 44 and 77 (d) 22 and 770
17. The HCF and LCM of two 2 digit numbers are 16 and 480, respectively. The numbers are
 (a) 40, 48 (b) 60, 72
 (c) 64, 80 (d) 80, 96
18. Find the greatest number which will exactly divide 200 and 320.
 (a) 10 (b) 20
 (c) 16 (d) 40
19. The greatest number which can divide 1356, 1868, 2764 leaving the same remainder in each case is
 (a) 260 (b) 64
 (c) 124 (d) 128
20. What is the least number which, when divided by 5, 6, 7 and 8 gives the remainder 3 but is divisible by 9?
 (a) 1463 (b) 1573
 (c) 1683 (d) 1793
21. The greatest number of four digits which when divided by 12, 16 and 24 leave remainder 2, 6 and 14 respectively is
 (a) 9974 (b) 9970
 (c) 9807 (d) 9998
22. What least number must be subtracted from 1936, so that the resulting number when divided by 9, 10 and 15 will leave in each case the same remainder 7?
 (a) 37 (b) 36
 (c) 39 (d) 30
23. The largest number, which divides 25, 73 and 97 to leave the same remainder in each case, is
 (a) 24 (b) 23
 (c) 21 (d) 6
24. The least multiple of 13, which on dividing by 4, 5, 6, 7 and 8 leaves remainder 2 in each case, is
 (a) 2520 (b) 842
 (c) 2522 (d) 840
25. What is the greatest number that will divide 307 and 330 leaving remainders 3 and 7, respectively?
 (a) 19 (b) 16
 (c) 17 (d) 23
26. Let the least number of six digits which when divided by 4, 6, 10 and 15 leaves in each case same remainder 2 be N. The sum of digits of N is
 (a) 3 (b) 5
 (c) 4 (d) 6
27. The smallest square number divisible by 10, 16 and 24 is:
 (a) 900 (b) 1600
 (c) 2500 (d) 3600
28. The greatest number, which divides 989 and 1327 leaving remainders 5 and 7 respectively, is
 (a) 8 (b) 16
 (c) 24 (d) 32
29. Find the largest number of four digits such that on dividing by 15, 18, 21 and 24 the remainders are 11, 14, 17 and 20 respectively.
 (a) 6557 (b) 7556
 (c) 5675 (d) 7664
30. The greatest number, that divides 122 and 243 leaving respectively, 2 and 3 as remainders is
 (a) 12 (b) 24
 (c) 30 (d) 120

Foundation

Solutions

1. (a); Given, fractions $\frac{2}{3}$, $\frac{4}{9}$ and $\frac{5}{6}$

$$\begin{aligned} \text{LCM of fractions} &= \frac{\text{LCM of numerators}}{\text{HCF of denominators}} \\ &= \frac{\text{LCM}(2, 4, 5)}{\text{HCF}(3, 9, 6)} = \frac{20}{3} \end{aligned}$$

2. (d); HCF of two numbers is 8. This means 8 is a factor common to both the numbers.

$\therefore$ LCM must be multiple of 8. By going through the options, 60 cannot be the LCM since it is not a multiple of 8.

3. (a); $\text{HCF} \times \text{LCM} = \text{Product of two numbers.}$

$$4 \times 520 = 52 \times \text{Second number}$$

$$\therefore \text{Second number} = \frac{4 \times 520}{52} = 40$$

4. (d); $\text{HCF} \times \text{LCM}$
 $= \text{First number} \times \text{Second number}$

$$\therefore \text{Second number} = \frac{96 \times 1296}{864} = 144$$

5. (d); $\text{HCF} \times \text{LCM} = \text{Product of two numbers}$.
 Then,

$$\text{LCM} = \frac{216}{6} = 36$$

6. (a); Let the numbers be $3x$ and $4x$

$$\text{HCF} = x = 4$$

$$\text{LCM of } 3x \text{ and } 4x = 12x = 12 \times 4 = 48$$

7. (c); Given, first number = 132

Since, each of the two numbers is a multiple of 12 (given),

$$\therefore \text{HCF} = 12 \text{ and } \text{LCM} = 1056 \quad (\text{given})$$

$$\text{LCM} \times \text{HCF} = \text{First number} \times \text{Second number}$$

$$\therefore \text{Second number} = \frac{1056 \times 12}{132} = 96$$

8. (d); Let the numbers be $3x$ and $4x$ respectively.

$$\text{HCF} \times \text{LCM}$$

$$= \text{First number} \times \text{Second number}$$

$$\therefore 2028 = 3x \times 4x$$

$$x^2 = \frac{2028}{12} = 169, \quad x = 13$$

$$\text{Numbers are } = 39, 52$$

$$\text{Sum of numbers} = 39 + 52 = 91$$

9. (c); Let the numbers be $3x$ and $4x$.

$$\text{LCM of } 3x \text{ and } 4x = 12x$$

$$\text{But given, LCM} = 240$$

$$\therefore 12x = 240$$

$$\Rightarrow x = \frac{240}{12} = 20$$

$$\therefore \text{Smaller number} = 3x = 3 \times 20 = 60$$

10. (b); $\text{HCF} \times \text{LCM} = \text{Product of two number}$

$$\therefore \text{Second number} = \frac{16 \times 160}{32} = 80$$

11. (b); $\text{HCF} \times \text{LCM} = \text{Product of two numbers}$

$$\therefore \text{LCM} = \frac{4107}{37} = 111, \quad ab \times \text{HCF} = \text{LCM}$$

$$\text{where } a, b \text{ are prime factors } ab = \frac{111}{37} = 3$$

$$\text{Prime number pairs } (3, 1)$$

$$\text{Numbers} = 3 \times \text{HCF}, 1 \times \text{HCF}$$

$$\therefore \text{Numbers are } 111 \text{ and } 37.$$

Hence, 111 is the greater number

12. (d); The ratio of two numbers is 3 : 4 and their HCF is 5. Their LCM is

Given that the numbers are in ratio of 3 : 4,

$$\text{First number} = 3 \times 5 = 15$$

$$\text{Second number} = 4 \times 5 = 20$$

$$\text{LCM} \times \text{HCF} = \text{Product of two numbers}$$

$$\therefore \text{LCM} = \frac{15 \times 20}{5} = 60$$

13. (c); We know that,

$$\text{LCM} \times \text{HCF} = \text{Product of two numbers}$$

$$\therefore \text{Second number} = \frac{26 \times 1820}{130} = 364$$

14. (a); We know that,

$$\text{HCF} \times \text{LCM} = \text{Product of two numbers}$$

$$\therefore \text{LCM} = \frac{12906 \times 14818}{478}$$

$$\text{LCM} = 400086$$

15. (b); Required number

$$= \text{HCF of } \{(110 - 2) \text{ and } (128 - 2)\}$$

$$= \text{HCF of } 108 \text{ and } 126$$

By Division Method

$$\begin{array}{r} 108 \overline{) 126} \{ 1 \\ \underline{108} \\ 18 \overline{) 108} \{ 6 \\ \underline{108} \\ 0 \end{array}$$

$$\therefore \text{Greatest number} = 18$$

16. (d); The LCM of 6, 12 and 18 = 36

36 is a perfect square of 6.

17. (a); The smallest number

$$= [\text{LCM of } (15, 20, 35) + k]$$

$$\begin{array}{r|l} 3 & 15, 20, 35 \\ 5 & 5, 20, 35 \\ \hline & 1, 4, 7 \end{array}$$

$$\text{LCM} = 4 \times 7 \times 5 \times 3 = 420$$

$$\text{Smallest number} = 420 + 8 = 428$$

18. (a); $30 \xrightarrow{(5)} 3 \times 2$
 $35 \xrightarrow{(5)} 7$

$$\text{So, HCF} = 5$$

19. (c); Let two numbers be $15x$ and $15y$. Where x and y are co-prime to each other.

$$\therefore 15x \times 15y = 6300$$

$$x \times y = \frac{6300}{15 \times 15}$$

$$x \times y = 28$$

Factors of 28 are 1, 2, 4, 7, 28

$$\therefore 28 = 1 \times 28 \text{ or } 4 \times 7$$

$$\therefore \text{There are only two possible pairs}$$

20. (a); $\text{HCF} + \text{LCM} = 403$

$$\text{HCF} + 12 \text{HCF} = 403$$

(It is given that $\text{LCM} = 12 \times \text{HCF}$)

$$13 \times \text{HCF} = 403$$

$$\text{HCF} = 31$$

$\therefore$ Product of two number is equal to product of their LCM and their HCF.

$$\therefore 93 \times \text{other number} = 31 \times 12 \times 31$$

$$\text{Other number} = \frac{31 \times 12 \times 31}{93} = 124$$

21. (b); We need to find the LCM of 12, 15, 20, 27.

$$12 = 2^2 \times 3$$

$$15 = 3 \times 5$$

$$20 = 2^2 \times 5$$

$$27 = 3^3$$

$$\begin{aligned} \text{LCM} &= \text{Product of highest powers of factors} \\ &= 2^2 \times 3^3 \times 5 = 540 \end{aligned}$$

22. (a); Let x be the remainder then $(25 - x)$, $(73 - x)$ and $(97 - x)$ will be exactly divisible by the required number.

$$\begin{aligned} \text{Required number} &= \text{HCF of } (73 - x) - (25 - x), (97 - x) - (73 - x) \text{ and } (97 - x) - (25 - x) \\ &= \text{HCF of } (73 - 25), (97 - 73) \text{ and } (97 - 25) \\ &= \text{HCF of } 48, 24 \text{ and } 72 = 24 \end{aligned}$$

23. (d); $\text{LCM of } 12, 15, 18, 27 = 540$

$$\text{Largest number of 5 digits} = 99999$$

$$\text{On dividing } 99999 \text{ by } 540, \text{ remainder} = 99$$

$$\therefore \text{Required number} = 99999 - 99 = 99900$$

24. (b); $20 - 14 = 25 - 19 = 35 - 29 = 40 - 34 = 6$

$$\begin{aligned} \text{Required number} &= (\text{LCM of } 20, 25, 35, 40) - 6 \\ &= 1400 - 6 = 1394 \end{aligned}$$

25. (a); To find the biggest measure, we have to find the HCF of 496, 403 and 713.

$$\text{HCF of } 496, 403 \text{ and } 713 = 31$$

26. (b); Required time = LCM of 20, 30 and 40

10	20, 30, 40
2	2, 3, 4
	1, 3, 2

$$\text{LCM} = 10 \times 2 \times 3 \times 2 = 120$$

Hence, the bells will simultaneously ring after 2h i.e., at 1 pm

27. (a); Required time = LCM of (200, 300, 360, 450) s.

10	200, 300, 360, 450
5	20, 30, 36, 45
3	4, 6, 36, 9
2	4, 2, 12, 3
2	2, 1, 6, 3
3	1, 1, 3, 3
	1, 1, 1, 1

$$\therefore \text{LCM} = 10 \times 5 \times 3 \times 2 \times 2 \times 3 = 1800 \text{ s}$$

28. (c); Let the numbers be ax and bx , where x is the HCF and $bx > ax$.

$$\therefore \text{LCM} = abx$$

$$abx = 2bx$$

$$\Rightarrow a = 2$$

$$\text{Again, } ax - x = 4$$

Putting the value of a , we get

$$2x - x = 4$$

$$\Rightarrow x = 4$$

$$\therefore \text{Smaller number} = ax = 2 \times 4 = 8$$

29. (a); All the 6 bells ring together will be LCM of (2, 4, 6, 8, 10 and 12)

2	2, 4, 6, 8, 10, 12
2	1, 2, 3, 4, 5, 6
3	1, 1, 3, 2, 5, 3
	1, 1, 1, 2, 5, 1

$\therefore$ They will ring together after

$$= 2 \times 2 \times 2 \times 3 \times 5 = 120 \text{ s}$$

i.e., they will ring together after 2 min

$\therefore$ Number of time they will ring together in 30

$$\text{min} = 1 + \frac{30}{2} = 1 + 15 = 16 \text{ times}$$

30. (c); Smallest number of boxes for buns alone

$$= \frac{\text{LCM of } 10 \text{ and } 8}{\text{Number of buns in a box}} = \frac{40}{10} = 4$$

Moderate

1. (a); Capacity of three containers containing diesel is 403 l, 434 l and 465 l, respectively.
Now, maximum capacity of the container that can measure the diesel of the three containers exactly
= HCF of quantity of three containers
= HCF (403, 434, 465)

$$\begin{array}{r} 403 \overline{)434} 1 \\ \underline{403} \\ 31 \overline{)403} 13 \\ \underline{403} \\ \times \end{array}$$

 Again, $31 \overline{)465} 15$

$$\begin{array}{r} 31 \overline{)465} 15 \\ \underline{465} \\ \times \end{array}$$

 So, Capacity of container = 31 L
2. (b); Time taken by A = $\frac{30}{5}$ h
Time taken by B = $\frac{30}{7}$ h
Time taken by C = $\frac{30}{8}$ h
Time taken by all to come together
= LCM of $\frac{30}{5}, \frac{30}{7}$ and $\frac{30}{8}$

$$= \frac{\text{LCM of } 30, 30 \text{ and } 30}{\text{HCF of } 5, 7 \text{ and } 8} = \frac{30}{1} = 30 \text{ h}$$
3. (b); LCM of 24, 36 and 54

2	24, 36, 54
2	12, 18, 27
2	6, 9, 27
3	3, 9, 27
3	1, 3, 9
3	1, 1, 3
	1, 1, 1

Required time = LCM of 24, 36 and 54 s = 216 s

$$= \frac{216}{60} = 3 \frac{36}{60} \text{ min}$$

$$= 3 \text{ min } 36 \text{ s}$$

$$\therefore \text{Time when they will change simultaneously}$$

$$= 10 : 15 : 00 + 3 \text{ min } 36 \text{ s}$$

$$= 10 : 18 : 36 \text{ am}$$
4. (c); Let the numbers be 48a and 48b, where a and b are coprimes,

$$\therefore 48a + 48b = 384$$

$$\Rightarrow 48(a + b) = 384$$

$$\Rightarrow a + b = \frac{384}{48} = 8 \quad \dots (i)$$

 Possible valid pairs of a and b satisfying this condition are (1, 7) and (3, 5).

$$\therefore \text{Numbers are } 48 \times 1 = 48$$

 and $48 \times 7 = 336$ or
 or, $48 \times 3 = 144$
 and $48 \times 5 = 240$

$$\therefore \text{Required difference} = 336 - 48 = 288$$

 or $240 - 144 = 96$
5. (c); Let the numbers be a and b.
According to the question,

$$a + b = 45 \quad \dots (i)$$

 Again, $a - b = \frac{1}{9}(a + b)$

$$\Rightarrow a - b = \frac{1}{9} \times 45 \Rightarrow a - b = 5 \quad \dots (ii)$$

 Adding equ. (i) and (ii), we get

$$2a = 50$$

$$a = \frac{50}{2} = 25$$

 On putting the value of a Eq. (i) we get

$$25 + b = 45$$

$$\Rightarrow b = 45 - 25 = 20,$$

$$a = 25 \text{ and } b = 20$$

$$\therefore \text{LCM of 25 and 20.}$$

5	25, 20
5	5, 4
	1, 4

$$\text{LCM} = 5 \times 5 \times 4 = 100$$
6. (c); Let the numbers be 4a and 4b where a and b are coprimes.

$$\text{LCM} = 4ab$$

$$\therefore \frac{(4a + 4b)}{4ab} = \frac{7}{12}$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} = \frac{7}{12} = \frac{1}{3} + \frac{1}{4}$$

$$\Rightarrow a = 3, b = 4$$

$$\therefore \text{First number} = (4 \times 3) = 12$$

$$\text{Second number} = (4 \times 4) = 16$$

$$\therefore \text{Smaller number} = 12$$

7. (b); Minimum number of students = LCM of 6, 8, 10.

$$\begin{array}{r|l} 2 & 6, 8, 10 \\ 2 & 3, 4, 5 \\ 2 & 3, 2, 5 \\ \hline & 3, 1, 5 \end{array}$$

$$\text{LCM} = 2 \times 2 \times 2 \times 3 \times 5 = 120$$

Hence, the minimum number of students = 120

8. (a); Let the number be $2x$, $3x$ and $4x$, respectively.

$$\therefore \text{HCF} = x = 12$$

$$\therefore \text{Numbers } 2 \times 12 = 24, 3 \times 12 = 36, 4 \times 12 = 48$$

LCM of 24, 36, 48

$$\begin{array}{r|l} 2 & 24, 36, 48 \\ 2 & 12, 18, 24 \\ 2 & 6, 9, 12 \\ 2 & 3, 9, 6 \\ 3 & 3, 9, 3 \\ 3 & 1, 3, 1 \\ \hline & 1, 1, 1 \end{array}$$

$$\therefore \text{LCM} = 2 \times 2 \times 2 \times 2 \times 3 \times 3 = 144$$

9. (d); Given,

$$3^{333} + 1 \text{ and } 3^{334} + 1 \text{ or } 27^{333} + 1^{333} \text{ and } 27^{334} + 1^{334}$$

Now, $x^m + a^m$ is divisible by $(x + a)$ when m is odd.

$27^{333} + 1^{333}$ is divisible by $(27 + 1) = 28$

Similarly, $27^{334} + 1^{334}$ is never divisible by $(x + a)$

So, the greatest common divisor between

$$(3^{333} + 1) \text{ and } (3^{334} + 1) \text{ is } 1.$$

10. (b); Maximum quantity in each can = HCF of (21, 42 and 63) $L = 21$ L

By Division Method

$$\begin{array}{r} 21 \overline{) 42} \begin{array}{l} 2 \\ \times \\ \hline 42 \end{array} \\ \hline 21 \overline{) 63} \begin{array}{l} 3 \\ \times \\ \hline 63 \end{array} \end{array}$$

$$\text{HCF} = 21 \text{ L}$$

$\therefore$ Least number of cans

$$= \frac{21}{21} + \frac{42}{21} + \frac{63}{21} = 1 + 2 + 3 = 6 \text{ cans.}$$

11. (c); To find the minimum number of rows, we determine the HCF of 24, 36 and 60.

$$\therefore \text{HCF of } 24, 36 \text{ and } 60 = 12$$

Thus, 12 fruits are there in a row.

$$\therefore \text{Number of rows} = \frac{24}{12} + \frac{36}{12} + \frac{60}{12} = 2 + 3 + 5 = 10$$

12. (c); Let the numbers be p , q and r which are coprime to one another.

$$\text{Now, } pq = 551 \text{ and } qr = 1073$$

$$q = \text{HCF of } 551 \text{ and } 1073$$

$$\begin{array}{r} 551 \overline{) 1073} \begin{array}{l} 1 \\ \times \\ \hline 551 \end{array} \\ \hline 522 \overline{) 551} \begin{array}{l} 1 \\ \times \\ \hline 522 \end{array} \\ \hline 29 \overline{) 522} \begin{array}{l} 18 \\ \times \\ \hline 522 \end{array} \end{array}$$

$$\therefore q = 29 \quad \therefore p = \frac{551}{29} = 19$$

$$\text{and } r = \frac{1073}{29} = 37$$

$\therefore$ Sum of three numbers.

$$= 19 + 29 + 37 = 85$$

13. (a); Number of books in each stack = HCF of (336, 240, 96)

$$\begin{array}{r} 240 \overline{) 336} \begin{array}{l} 1 \\ \times \\ \hline 240 \end{array} \\ \hline 96 \overline{) 240} \begin{array}{l} 2 \\ \times \\ \hline 192 \end{array} \\ \hline 48 \overline{) 96} \begin{array}{l} 2 \\ \times \\ \hline 96 \end{array} \end{array}$$

$\therefore$ Number of books in each stack = 48

$$\therefore \text{Total number of stacks} = \frac{336}{48} + \frac{240}{48} + \frac{96}{48} = 7 + 5 + 2 = 14$$

14. (c); Given, HCF = 3, LCM = 105

Now, let the numbers be $3a$ and $3b$,

$$\therefore 3a + 3b = 36$$

$$\Rightarrow a + b = 12 \quad \dots (i)$$

$$\text{and LCM } 3ab = 105 \quad \dots (ii)$$

Dividing Eq. (i) by Eq. (ii), we have

$$\frac{a}{3ab} + \frac{b}{3ab} = \frac{12}{105}$$

$$\Rightarrow \frac{1}{3a} + \frac{1}{3b} = \frac{4}{35} \Rightarrow \frac{1}{a} + \frac{1}{b} = \frac{12}{35}$$

15. (d); Length of the floor = 15 m 17 cm = 1517 cm

$$\text{Breadth of the floor} = 9 \text{ m } 2 \text{ cm} = 902 \text{ cm}$$

$$\text{Area of the floor} = 1517 \times 902 \text{ cm}^2$$

The number of square tiles will be least, when the size of each tile is maximum.

$$\therefore \text{Size of each tile} = \text{HCF of } 1517 \text{ and } 902$$

$$\begin{array}{r}
 902 \overline{)1517} (1 \\
 \underline{902} \\
 615 \overline{)902} (1 \\
 \underline{615} \\
 287 \overline{)615} (2 \\
 \underline{574} \\
 41 \overline{)287} (7 \\
 \underline{287} \\
 \times
 \end{array}$$

HCF = 41

∴ Required number of tiles

$$= \frac{\text{Area of the floor}}{\text{Area of each square tile}} = \frac{1517 \times 902}{41 \times 41} = 814$$

16. (d); Given, HCF of a and b = 12

Let the numbers be 12x and 12y, where x and y are coprime.

But given $a > b > 12$

Smallest coprime pair for the above condition = (3, 2)

$$\therefore a = 36 \text{ and } b = 24$$

17. (b); Given,

$$P = 2^3 \times 3^{10} \times 5, \quad Q = 2^5 \times 3 \times 7$$

[∵ 2^3 and 3 is common factor to both P and Q]

$$\text{HCF} = 2^3 \times 3$$

18. (b); HCF = 12

Numbers = 12a and 12b where, a and b are coprimes

$$\therefore 12a + 12b = 84 \Rightarrow 12(a + b) = 84$$

$$\Rightarrow a + b = \frac{84}{12} = 7$$

∴ Possible pairs of numbers satisfying this condition = (1, 6), (2, 5) and (3, 4)

Hence, pairs of required numbers = 3.

19. (c); HCF of two numbers = 4

Hence, the numbers can be expressed as 4a and 4b, where a and b are coprime,

$$4a + 4b = 36, \quad a + b = 9$$

Now, possible pairs satisfying above condition are (1, 8), (4, 5), (2, 7).

∴ 3 pairs are possible

20. (a); We know that,

HCF × LCM = Product of two numbers

$$\therefore \text{LCM} = \frac{12908 \times 14808}{672} = 284437$$

21. (c); Factors of 50 = $5^2 \times 2$

$$\text{Factors of } 80 = 5^1 \times 2^4$$

$$\text{H.C.F. of } 50 \text{ \& } 80 = 5^1 \times 2^1 = 10$$

The capacity of the mug must be 10 l

22. (a); LCM of 48, 72 and 108 = 432

The traffic lights will change simultaneously after 432 seconds or 7 m 12 secs.

∴ They will change simultaneously at

$$= 8 : 20 \text{ hours} + 7 \text{ m} + 12 \text{ sec.} = 8 : 27 : 12 \text{ hrs.}$$

23. (b); Product of two numbers = H.C.F. × L.C.M.

$$7 \times 30 \times \text{Second number} = 30 \times 2310$$

$$\text{Second number} = \frac{30 \times 2310}{7 \times 30} = 330$$

24. (b); LCM of 2, 3, 4, 5 and 6 = 60

Other numbers divisible by 2, 3, 4, 5, 6 are 60k, where k is a positive integer. Since $2 - 1 = 1$, $3 - 2 = 1$, $4 - 3 = 1$, $5 - 4 = 1$ and $6 - 5 = 1$, the remainder in each case is less than the divisor by 1. Now, the required number is to be divisible by 7. Hence, we must choose the least value of k which will make $(60k - 1)$ divisible by 7. Putting k equal to 1, 2, 3 etc. in succession, we find that k should be 2

∴ The required number = $60k - 1$

$$= 60 \times 2 - 1 = 119$$

25. (d); Required number

$$= (\text{LCM of } 12, 15, 20 \text{ \& } 54) + 4$$

$$= 540 + 4 = 544$$

26. (a); Required number of students = HCF of 1001 and 910

$$\begin{array}{r}
 910 \overline{)1001} (1 \\
 \underline{910} \\
 91 \overline{)910} (10 \\
 \underline{910} \\
 0
 \end{array}$$

Hence, HCF = 91

27. (d); Required time = LCM of 252, 308 and 198 s.

$$\begin{array}{r|l}
 2 & 252, 308, 198 \\
 \hline
 2 & 126, 154, 99 \\
 \hline
 7 & 63, 77, 99 \\
 \hline
 9 & 9, 11, 99 \\
 \hline
 11 & 1, 11, 11 \\
 \hline
 & 1, 1, 1
 \end{array}$$

$$\therefore \text{LCM} = 2 \times 2 \times 7 \times 9 \times 11$$

$$= 2772 \text{ s} = \frac{2772}{60}$$

$$= 46 \frac{1}{5} \text{ min} = 46 \text{ min } 12 \text{ s}$$

28. (c); Minimum distance each should cover, so that all can cover the distance in complete steps

$$= \text{LCM of } (63, 70, 77) = 6930 \text{ cm}$$

29. (b); LCM of 15, 18, 21, 24

2	15, 18, 21, 24
2	15, 9, 21, 12
2	15, 9, 21, 6
3	15, 9, 21, 3
3	5, 3, 7, 1
5	5, 1, 7, 1
7	1, 1, 7, 1
	1, 1, 1, 1

$$\text{LCM} = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7 = 2520$$

Largest number of four digit = 9999

$$\begin{array}{r} 2520 \overline{) 9999} 3 \\ \underline{7560} \end{array}$$

$$2439$$

$$\text{Required number} = 9999 - 2439 - 4 = 7556$$

$$\text{Where, } 4 = \begin{cases} 15 - 11 = 4 \text{ or} \\ 18 - 14 = 4 \text{ or} \\ 21 - 17 = 4 \text{ or} \\ 24 - 20 = 4 \end{cases}$$

30. (b); LCM of 9 and 6 = 18

Total numbers from 1 to 200 divisible by 18 = 11

Total numbers from 1 to 100 divisible by 18 = 5

$\therefore$ Required numbers from 100 to 200
divisible by 18 = 11 - 5 = 6

Difficult

1. (b); The largest possible length of the tape = HCF of 525, 1050, 1155 = 105

Hence (b) is the correct answer.

2. (c); The maximum capacity of the vessel = HCF of 1653, 2261 and 2527 = 19

Hence, (c) is the correct option.

3. (b); Minimum number of rows means max. number of trees per row, also equal number of trees per row is required so we need to find the HCF of 36, 144 and 234 to find the maximum number of trees in a row.

Thus HCF of 36, 144 and 234 = 18

$$\text{Thus the number of rows} = \frac{\text{Total no. of trees}}{\text{No. of trees in a row}} = \frac{36 + 144 + 234}{18} = 23$$

Hence (b) is correct answer.

4. (a); The required number must be divisible by the given numbers so it can be the LCM or its multiple number.

Now the LCM of 6, 40, 49 and 75

$$= 2 \times 2 \times 2 \times 3 \times 5 \times 5 \times 7 \times 7$$

But the required number is a perfect square

Thus the LCM must be multiplied by $2 \times 3 = 6$.

Thus the required number

$$= (2 \times 2 \times 2 \times 3 \times 5 \times 5 \times 7 \times 7) \times (2 \times 3)$$

$$= 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 5 \times 7 \times 7$$

$$= 176400$$

5. (b); The three bells toll together only at the LCM of the times they toll individually.

Thus the LCM of 48, 72 and 108 is 432 seconds.

Therefore all the bells will toll together at 6 : 07 : 12 AM

($\because$ 432 seconds = 7 minutes 12 seconds)

6. (c); Since the difference between the divisors and the respective remainders is same.

Hence the least possible number

$$= (\text{LCM of } 18, 35 \text{ and } 42) - 16$$

$$= 630 - 16 \quad [\because (18 - 2) = (35 - 19) = (42 - 26) = 16]$$

$$= 614$$

7. (a); $0.0005 \Rightarrow 5$

$$0.0050 \Rightarrow 50$$

$$0.1500 \Rightarrow 1500$$

$$0.1750 \Rightarrow 1750$$

$$0.5000 \Rightarrow 5000$$

$$3.5000 \Rightarrow 35000$$

Then the HCF of 5, 50, 1500, 1750, 5000 and 35000 is 5.

So the HCF of the given number is 0.0005 (since there are four digits in all the adjusted (or equated) decimal places.)

8. (b); The required number = (LCM of 2, 3, 4, 5, 6) $K + 1 = 7l = 60K + 1 = 7l$

$$\Rightarrow \frac{60k + 1}{7} = l$$

Now put the least possible value of k such that l must be a positive integer. Hence at $k = 5$, l is an integer. Thus, the required value is $60 \times 5 + 1 = 301$.

9. (b); The required time = LCM of 36, 40 and 48

$$= 720 \text{ seconds} = 12 \text{ minutes}$$

Hence, (b) is the right choice.

10. (c); Minimum number of corks =

$$\frac{403 + 434 + 465}{\text{HCF of } (403, 434, 465)} = \frac{1302}{31} = 42$$

11. (c); Time taken for each of three persons is

$$\text{respectively } \frac{6}{3}, \frac{6}{2\frac{1}{2}} \text{ and } \frac{6}{1\frac{1}{4}} \text{ hrs}$$

i.e., $\frac{2}{1}$, $\frac{12}{5}$ and $\frac{24}{5}$ hrs.

So, it is required to find the LCM of

$$\frac{2}{1}, \frac{12}{5}, \frac{24}{5} = \frac{24}{1} = 24 \text{ hr}$$

Hence, (c) is the right choice.

12. (b); Let LCM be x and HCF be y .

According to the question,

$$\text{LCM} = 4 \times \text{HCF}$$

$$\Rightarrow x = 4y$$

According to the question,

$$\text{LCM} + \text{HCF} = 125$$

$$x + y = 125$$

Putting the value of x , we get

$$5y = 125$$

$$\Rightarrow y = 25$$

$$\therefore \text{HCF} = 25 \text{ and } \text{LCM} = 4 \times 25 = 100$$

We know that, $\text{HCF} \times \text{LCM}$

$$= \text{First number} \times \text{Second number}$$

$$\text{Second number} = \frac{\text{HCF} \times \text{LCM}}{\text{First number}}$$

$$= \frac{100 \times 25}{100} = 25$$

13. (d); Let the numbers be $10a$ and $10b$, where a and b are coprime

$$\therefore \text{LCM of } 10a \text{ and } 10b = 10ab$$

$$\Rightarrow 10ab = 120 \Rightarrow ab = 12$$

Possible pairs = (3, 4) or (1, 12)

Sum of the numbers :

$$(a, b) = (3, 4) :$$

$$(3 \times 10) + (4 \times 10) = 30 + 40 = 70$$

$$(a, b) = (1, 12) :$$

$$(1 \times 10) + (12 \times 10)$$

$$= 10 + 120 = 130$$

14. (d); Let the original fraction be $\frac{x}{y}$.

According to the question,

$$\frac{x-4}{y+1} = \frac{1}{6} \Rightarrow 6x - 24 = y + 1$$

$$\Rightarrow 6x - y = 25$$

Again, according to the question,

$$\frac{x+2}{y+1} = \frac{1}{3}$$

$$\Rightarrow 3x + 6 = y + 1$$

$$\Rightarrow 3x - y = -5$$

... (i)

... (ii)

On subtracting Eq. (ii) from Eq. (i), we get

$$6x - y = 25$$

$$\frac{3x - y = -5}{3x = 30}$$

$$x = \frac{30}{3} = 10$$

on putting the value of x in Eq. (i), we get

$$6 \times 10 - y = 25$$

$$\Rightarrow -y = 25 - 60 \Rightarrow -y = -35$$

$$\Rightarrow y = 35$$

$$\therefore x = 10 \text{ and } y = 35$$

LCM of 10 and 35

$$\begin{array}{r|l} 5 & 10, 35 \\ \hline & 2, 7 \end{array}$$

$$\text{LCM} = 2 \times 7 \times 5 = 70$$

15. (c); A makes one complete round in $5/(5/2)$

$$= 2 \text{ h} \quad \left(\because \text{Speed} = \frac{\text{Distance}}{\text{Time}} \right)$$

B completes in $\frac{5}{3}$ h and C completes in $\frac{5}{2}$ h.

Hence, the required time

$$= \text{LCM of } \left(2, \frac{5}{3} \text{ and } \frac{5}{2} \right) \text{ h}$$

$$= \frac{\text{LCM of } 2, 5, 5}{\text{HCF of } 3, 2} = \frac{10}{1} = 10 \text{ h}$$

Hence, A, B and C will meet after 10 h.

16. (b); By the Technique

$$\text{First number} \times \text{Second number} = \text{HCF} \times \text{LCM}$$

$$\therefore \text{Second number} = \frac{15 \times 300}{60} = 75$$

17. (c); Let the numbers be $7a$ and $7b$, where a and b are coprime. Now, $\text{LCM of } 7a \text{ and } 7b = 7ab$

$$\therefore 7ab = 140$$

$$ab = \frac{140}{7} = 20$$

Now, required values of a and b whose product is 20 are 4 and 5.

Numbers are 28 and 35 and they lie between 20 and 45.

$$\text{Sum of the numbers} = 28 + 35 = 63$$

18. (c); We know that

Numbers divisible by 4 and 6 will be multiples of the LCM of 4 and 6 i.e., 12

Now, numbers from 1 to 600 divisible by

$$12 = \frac{600}{12} - 1 = 49$$

(minus 1 because 600 is excluded)

Now, numbers divisible by 12 from 1 to 100

$$= \frac{100}{12} = 8$$

$$\therefore \text{Numbers divisible by between 100 and 600} \\ = 49 - 8 = 41$$

$$\begin{array}{r|l} 19. (a); & 2 \mid 18, 27, 36 \\ & 2 \mid 9, 27, 18 \\ & 3 \mid 9, 27, 9 \\ & 3 \mid 3, 9, 3 \\ & 3 \mid 1, 3, 1 \\ & 1, 1, 1 \end{array}$$

$$\text{LCM} = 2 \times 2 \times 3 \times 3 \times 3 = 108$$

Required number

$$= [\text{LCM of } (18, 27, 36)] - k = 108 - 13 = 95$$

{where, $k = 18 - 5$ or $27 - 14$ or $36 - 23 = 13$ }

20. (b); LCM of 16, 24, 30 and 36.

$$\begin{array}{r|l} & 2 \mid 16, 24, 30, 36 \\ & 2 \mid 8, 12, 15, 18 \\ & 2 \mid 4, 6, 15, 9 \\ & 2 \mid 2, 3, 15, 9 \\ & 3 \mid 1, 3, 15, 9 \\ & 3 \mid 1, 1, 5, 3 \\ & 5 \mid 1, 1, 5, 1 \\ & 1, 1, 1, 1 \end{array}$$

$$\text{LCM} = 2 \times 2 \times 2 \times 3 \times 2 \times 5 \times 3 = 720$$

We know that largest five digit number is 99999

$$\begin{array}{r} 720 \overline{) 99999} \\ \underline{720} \\ 2799 \\ \underline{2160} \\ 6399 \\ \underline{5760} \\ 639 \end{array}$$

$$\text{Required number} = (99999 - 639) + 10 = 99370$$

Previous Year (Memory Based)

1. (c); Here, first number $= 2 \times 44 = 88$,
HCF $= 44$ and LCM $= 264$.
By the formula
1st number $\times$ 2nd number $= \text{HCF} \times \text{LCM}$
 $\Rightarrow 88 \times \text{2nd number} = 44 \times 264$

$$\Rightarrow \text{2nd number} = \frac{264 \times 44}{88} = 132$$

$$\Rightarrow \text{2nd number} = 132$$

2. (b); Let the common ratio $= x$
Then, numbers are $5x$ and $6x$
Now, HCF of these two numbers is x
By the technique
 $\text{LCM} \times \text{HCF} = \text{Product of two numbers}$
 $\Rightarrow 480 \times x = 5x \times 6x \Rightarrow 480x = 30x^2$
 $\therefore x = 16 \quad \therefore \text{HCF is } 16$

3. (d); We know that,
LCM of two coprimes is equal to their product,
Hence, LCM $= 117$

4. (c); If the numbers be $2x$ and $3x$, then LCM of $2x$ and $3x = 6x$
LCM $= 48$

$$\therefore 6x = 48 \Rightarrow x = \frac{48}{6} = 8$$

$\therefore$ The numbers are $(8 \times 2 = 16)$
and $(8 \times 3 = 24)$, respectively.

$$\therefore \text{Sum} = 16 + 24 = 40$$

5. (d); Let the numbers be $4x$ and $5x$.

$$\therefore \text{HCF} = 8 = x$$

$$\text{First number} = 8 \times 4 = 32$$

$$\text{Second number} = 8 \times 5 = 40$$

$$\therefore \text{LCM of } 32, 40 = 160$$

6. (b); Let the number be $2a$ and $2b$, where a and b are coprime

$$\therefore \text{LCM} = 2ab$$

$$2ab = 84$$

$$ab = 42 = 6 \times 7$$

$$\therefore \text{Numbers are } 12 \text{ and } 14.$$

$$\therefore \text{Sum } 12 + 14 = 26$$

7. (d); By the technique

$$\text{HCF} \times \text{LCM} = \text{First number} \times \text{Second number}$$

$$\therefore \text{Second number} = \frac{8 \times 48}{24} = 16$$

8. (c); Let the larger number be a .

$$\text{Smaller number} = a - 2$$

$$\text{HCF} \times \text{LCM} = \text{Product of two numbers}$$

$$24 = a(a - 2)$$

$$\Rightarrow a^2 - 2a - 24 = 0$$

$$\Rightarrow a^2 - 6a + 4a - 24 = 0$$

$$\Rightarrow a(a - 6) + 4(a - 6) = 0$$

$$\Rightarrow a = 6, -4$$

$$\text{But } a \neq -4 \quad \therefore a = 6$$

9. (c); Let the numbers are ax and bx

where $x = \text{HCF} = 15$

ATQ,

$$ax \times bx = 6300$$

$$ab = \frac{6300}{225} = 28$$

$\therefore$ possible pairs $\Rightarrow 28 \times 1, 7 \times 4$

Hence there are two pairs.

10. (d); Let LCM be x and HCF be y .

According to the question.

$$\text{LCM} = 20 \times \text{HCF}$$

$$\text{i.e., } x = 20y$$

$$\text{and } x + y = 2520$$

Putting the value of x , we get

$$20y + y = 2520$$

$$\Rightarrow 21y = 2520$$

$$\Rightarrow y = \frac{2520}{21} = 120$$

$$\therefore \text{LCM} = x = 120 \times 20 = 2400$$

$$\text{LCM} \times \text{HCF} = \text{Product of two numbers}$$

$$2400 \times 120 = 48 \times x, \quad x = 600$$

11. (b); Given fractions = $\frac{2}{4}, \frac{4}{5}$ and $\frac{6}{7}$

$$\text{HCF of fractions} = \frac{\text{HCF of numerators}}{\text{LCM of denominators}}$$

$$= \frac{\text{HCF}(2, 4, 6)}{\text{LCM}(3, 5, 7)} = \frac{2}{105}$$

12. (a); LCM of 28 and 42

$$\begin{array}{r|l} 2 & 28, 42 \\ 2 & 14, 21 \\ 7 & 7, 21 \\ \hline & 1, 3 \end{array}$$

$$\therefore \text{LCM} = 2 \times 2 \times 7 \times 3 = 84$$

$$\text{HCF of 28 and 42}$$

By Division method

$$\begin{array}{r} 28 \overline{) 42} 1 \\ \underline{28} \\ 14 \overline{) 28} 2 \\ \underline{28} \\ 0 \end{array}$$

$$\therefore \text{HCF} = 14$$

$$\therefore \text{Ratio} = \frac{\text{LCM}}{\text{HCF}} = \frac{84}{14} = \frac{6}{1} = 6:1$$

13. (a); $\text{LCM} = (x^2 + 6x + 8)(x + 1)$

$$\text{or } (x + 4)(x + 2)(x + 1)$$

$$\text{HCF} = (x + 1)$$

$$1\text{st expression} = x^2 + 3x + 2$$

$$\text{or } (x + 1)(x + 2)$$

As we know that,

$$\text{product of two expressions} = \text{LCM} \times \text{HCF}$$

$$(x + 1)(x + 2) \times 2\text{nd expression}$$

$$= (x + 4)(x + 2)(x + 1)(x + 1)$$

$$2\text{nd expressions}$$

$$= \frac{(x + 4)(x + 2)(x + 1)(x + 1)}{(x + 1)(x + 2)}$$

$$= (x + 4)(x + 1) = x^2 + 4x + x + 4 = x^2 + 5x + 4$$

14. (c); Let the numbers be $3x$ and $4x$.

$$\text{LCM of } 3x \text{ and } 4x = 12x$$

$$\text{Now, LCM} = 84$$

$$\text{Then, } 12x = 84$$

$$\Rightarrow x = \frac{84}{12} = 7$$

$$\therefore \text{Greatest number} = 4x = 4 \times 7 = 28$$

15. (b); Let the number be $29a$ and $29b$, respectively where a and b are coprimes

$$\text{LCM of } 29a \text{ and } 29b = 29ab$$

$$\text{Now, } 29ab = 4147$$

$$\therefore ab = \frac{4147}{29} = 143$$

$$\text{Thus, } ab = 11 \times 13$$

$$\text{First number} = (29 \times 11) = 319$$

$$\text{Second number} = (29 \times 13) = 377$$

$$\therefore \text{Sum of numbers} = 319 + 377 = 696$$

16. (b); Factors of 11 and 385 are

$$11 = 11 \times 1, \quad 385 = 11 \times 5 \times 7$$

$$\therefore \text{LCM} = 11 \times 5 \times 7 = 385$$

$$\text{HCF} = 11$$

$$\text{First number} = 11 \times 5 = 55$$

$$\text{Second number} = 11 \times 7 = 77$$

$$\Rightarrow (11, 385) \text{ or } (55, 77)$$

17. (d); HCF of the two digit numbers = 16

Hence, let the numbers be $16a$ and $16b$.

where, a and b are coprimes.

$$\text{Now, HCF} \times \text{LCM} = \text{Product of two numbers.}$$

$$\Rightarrow 16a \times 16b = 16 \times 480$$

$$\Rightarrow ab = \frac{16 \times 480}{16 \times 16} = 30$$

Possible pairs of a and b satisfying the condition $ab = 30$ are $(3, 10), (5, 6), (1, 30), (2, 15)$. Since the

numbers are of 2 digit each.

Hence, required pair is (5, 6).

First number = $16 \times 5 = 80$

Second number = $16 \times 6 = 96$

18. (d); Greatest number that can exactly divide 200 and 320 = HCF of 200 and 320 = 40

$$\begin{array}{r} 200 \overline{)320} 1 \\ \underline{200} \\ 120 \\ \underline{120} \\ 0 \\ \underline{0} \\ 0 \\ \underline{0} \\ 0 \\ \underline{0} \\ 0 \end{array}$$

Hence the greatest number is 40.

19. (d); Required number = HCF of $\{(1868-1356), (2764-1868), (2764-1356)\}$
= HCF of (512, 896, 1408)

$$\begin{array}{r} 512 \overline{)896} 1 \\ \underline{512} \\ 384 \\ \underline{384} \\ 0 \\ \underline{0} \\ 0 \end{array}$$

Hence, required number is 128.

20. (c); LCM of 5, 6, 7, 8 = $35 \times 24 = 840$
Required number = $840x + 3$, such that it is exactly divisible by 9.

By hit and Trial

for $x = 2$, it is divisible by 9.

Required number = $840x + 3 = 840 \times 2 + 3 = 1683$
(these type of questions can be solved with the help of given options)

(Out of all the given options, only 1683 is divisible by 9.)

21. (a); LCM of 12, 16 and 24

$$\begin{array}{l} 2 \mid 12, 16, 24 \\ 2 \mid 6, 8, 12 \\ 3 \mid 3, 4, 6 \\ 2 \mid 1, 4, 2 \\ 2 \mid 1, 2, 1 \\ 1, 1, 1 \end{array}$$

$$= 2 \times 2 \times 2 \times 2 \times 3 = 48$$

$$\begin{array}{r} 48 \overline{)9999} 208 \\ \underline{96} \\ 399 \\ \underline{384} \\ 15 \end{array}$$

$\therefore$ Greatest four digit numbers divisible by 48

$$9999 - 15 = 9984$$

$\therefore$ Required number = $9984 - 10 = 9974$

(10 is the difference of each remainder)

22. (c); LCM of 9, 10 and 15

$$\begin{array}{l} 2 \mid 9, 10, 15 \\ 3 \mid 9, 5, 15 \\ 3 \mid 3, 5, 5 \\ 5 \mid 1, 5, 5 \\ 1, 1, 1 \end{array}$$

$$\therefore \text{LCM} = 2 \times 3 \times 3 \times 5 = 90$$

$$\begin{array}{r} 90 \overline{)1936} 21 \\ \underline{180} \\ 136 \\ \underline{90} \\ 46 \end{array}$$

$$\therefore \text{Required number} = 46 - 7 = 39$$

23. (a); Required number
= HCF of $[|25 - 73|, |73 - 97|, |97 - 25|]$
= HCF of {48, 24, 72}
HCF = $2 \times 2 \times 2 \times 3 = 24$

$$\therefore \text{HCF} = 24$$

$$\therefore \text{Largest number} = 24$$

24. (c); LCM of 4, 5, 6, 7 and 8 = 840

$$\text{Required number} = 840x + 2$$

By Hit and Trial

Putting $x = 3$

$$\text{we get} = 840x + 2 = 840 \times 3 + 2 = 2522$$

2522 is a multiple of 13.

25. (a); Greatest number
= HCF of $[(307 - 3), (330 - 7)]$
= HCF of (304, 323)

$$\begin{array}{r} 304 \overline{)323} 1 \\ \underline{304} \\ 19 \\ \underline{19} \\ 0 \end{array}$$

$$\therefore \text{Required number} = 19$$

26. (b); Least six digit number is 100000

LCM of 4, 6, 10, 15

$$\begin{array}{l} 2 \mid 4, 6, 10, 15 \\ 2 \mid 2, 3, 5, 15 \\ 3 \mid 1, 3, 5, 15 \\ 5 \mid 1, 1, 5, 5 \\ 1, 1, 1, 1 \end{array}$$

$$\therefore \text{LCM} = 2 \times 2 \times 3 \times 5 = 60$$

$$\begin{array}{r} 60 \overline{)100000} 1666 \\ \underline{60} \\ 400 \\ \underline{360} \\ 400 \\ \underline{360} \\ 400 \\ \underline{360} \\ 40 \end{array}$$

$$\therefore \text{Required number}$$

$$= 100000 + (60 - 40) + 2 = 100022$$

∴ Sum of the digits of

$$N = 1 + 0 + 0 + 0 + 2 + 2 = 5$$

27. (d); LCM of 10, 16, 24

$$\begin{array}{r|l} 2 & 10, 16, 24 \\ \hline 2 & 5, 8, 12 \\ \hline 2 & 5, 4, 6 \\ \hline & 5, 2, 3 \end{array}$$

∴ LCM of $2^2 \times 2^2 \times 5 \times 3$

[∵ powers must be equal for number to be perfect square]

∴ Required number

$$= 2^2 \times 2^2 \times 5^2 \times 3^2 = 4 \times 4 \times 25 \times 9 = 3600$$

28. (c); By the technique

Required number

$$= \text{HCF of } [(989 - 5), (1327 - 7)]$$

$$= \text{HCF of } (984, 1320)$$

$$\begin{array}{r} 984 \overline{)1320} 1 \\ \underline{984} \\ 336 \\ 336 \overline{)984} 2 \\ \underline{672} \\ 312 \\ 312 \overline{)336} 1 \\ \underline{312} \\ 24 \\ 24 \overline{)312} 13 \\ \underline{312} \\ \hline \times \end{array}$$

∴ HCF = 24

∴ Required number = 24

29. (b); LCM of 15, 18, 21, 24

$$\begin{array}{r|l} 2 & 15, 18, 21, 24 \\ \hline 2 & 15, 9, 21, 12 \\ \hline 2 & 15, 9, 21, 6 \\ \hline 3 & 15, 9, 21, 3 \\ \hline 3 & 5, 3, 7, 1 \\ \hline 5 & 5, 1, 7, 1 \\ \hline 7 & 1, 1, 7, 1 \\ \hline & 1, 1, 1, 1 \end{array}$$

$$\text{LCM} = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7 = 2520$$

Largest number of four digit = 9999

$$\begin{array}{r} 2520 \overline{)9999} 3 \\ \underline{7560} \\ 2439 \end{array}$$

$$\text{Required number} = 9999 - 2439 - 4 = 7556$$

$$\text{where, } 4 = \begin{cases} 15 - 11 = 4 \text{ or} \\ 18 - 14 = 4 \text{ or} \\ 21 - 17 = 4 \text{ or} \\ 21 - 20 = 4 \end{cases}$$

30. (d); Required number = HCF of [(122-2), (243-3)]

i.e., HCF of (120, 240)

By Division Method

$$\begin{array}{r} 120 \overline{)240} 2 \\ \underline{240} \\ \hline \times \end{array}$$

∴ Required number = 120